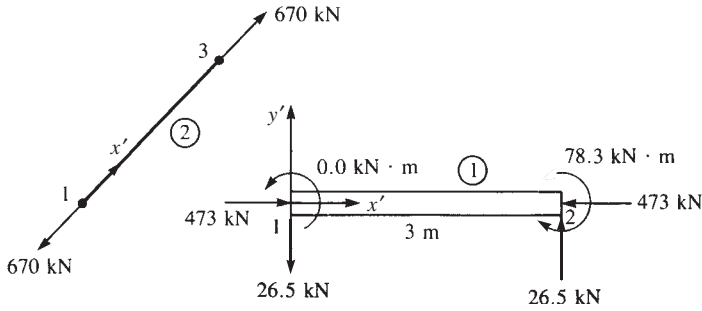




■ **Figure 5-11** Free-body diagrams of the bar (element 2) and beam (element 1) elements of Figure 5-10

Similarly, using the lower left 3×3 part of Eq. (5.1.6), we have at node 2,

$$\begin{Bmatrix} f'_{2x} \\ f'_{2y} \\ m'_2 \end{Bmatrix} = 70 \times 10^3 \begin{bmatrix} -2 & 0 & 0 \\ 0 & -0.067 & -0.10 \\ 0 & 0.10 & 0.10 \end{bmatrix} \begin{Bmatrix} 0.00338 \\ -0.0225 \\ 0.0113 \end{Bmatrix}$$

The matrix product then yields

$$f'_{2x} = -473 \text{ kN} \quad f'_{2y} = 26.5 \text{ kN} \quad m'_2 = -78.3 \text{ kN} \cdot \text{m} \quad (5.2.53)$$

To help interpret the results of Eqs. (5.2.49), (5.2.50), (5.2.52), and (5.2.53), freebody diagrams of the bar and beam elements are shown in Figure 5-11. To further verify the results, we can show a check on equilibrium of node 1 to be satisfied. You should also verify that moment equilibrium is satisfied in the beam.

5.3 Inclined or Skewed Supports—Frame Element

For the frame element with inclined support at node 3 in Figure 5-12, the transformation matrix $[T]$ used to transform global to local nodal displacements is given by Eq. (5.1.10).

In the example shown in Figure 5-12, we use $[T]$ applied to node 3 as follows:

$$\begin{Bmatrix} u'_3 \\ v'_3 \\ \phi'_3 \end{Bmatrix} = \begin{bmatrix} \cos \alpha & \sin \alpha & 0 \\ -\sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} u_3 \\ v_3 \\ \phi_3 \end{Bmatrix} \quad (5.3.1)$$

The same steps as given in Section 3.9 then follow for the plane frame. The resulting equations for the plane frame in Figure 5-12 are [see also Eq. (3.9.13)]

$$[T_i]\{f\} = [T_i][K][T_i]^T \{d\} \quad (5.3.2)$$